

GEOSPATIAL REPRESENTATION OF RADIONUCLIDE DISPERSION AND ASSOCIATED EFFECTIVE DOSE MODELED WITH ARTM VIEW: A CASE STUDY AT CDTN'S RADIOPHARMACEUTICAL UNIT



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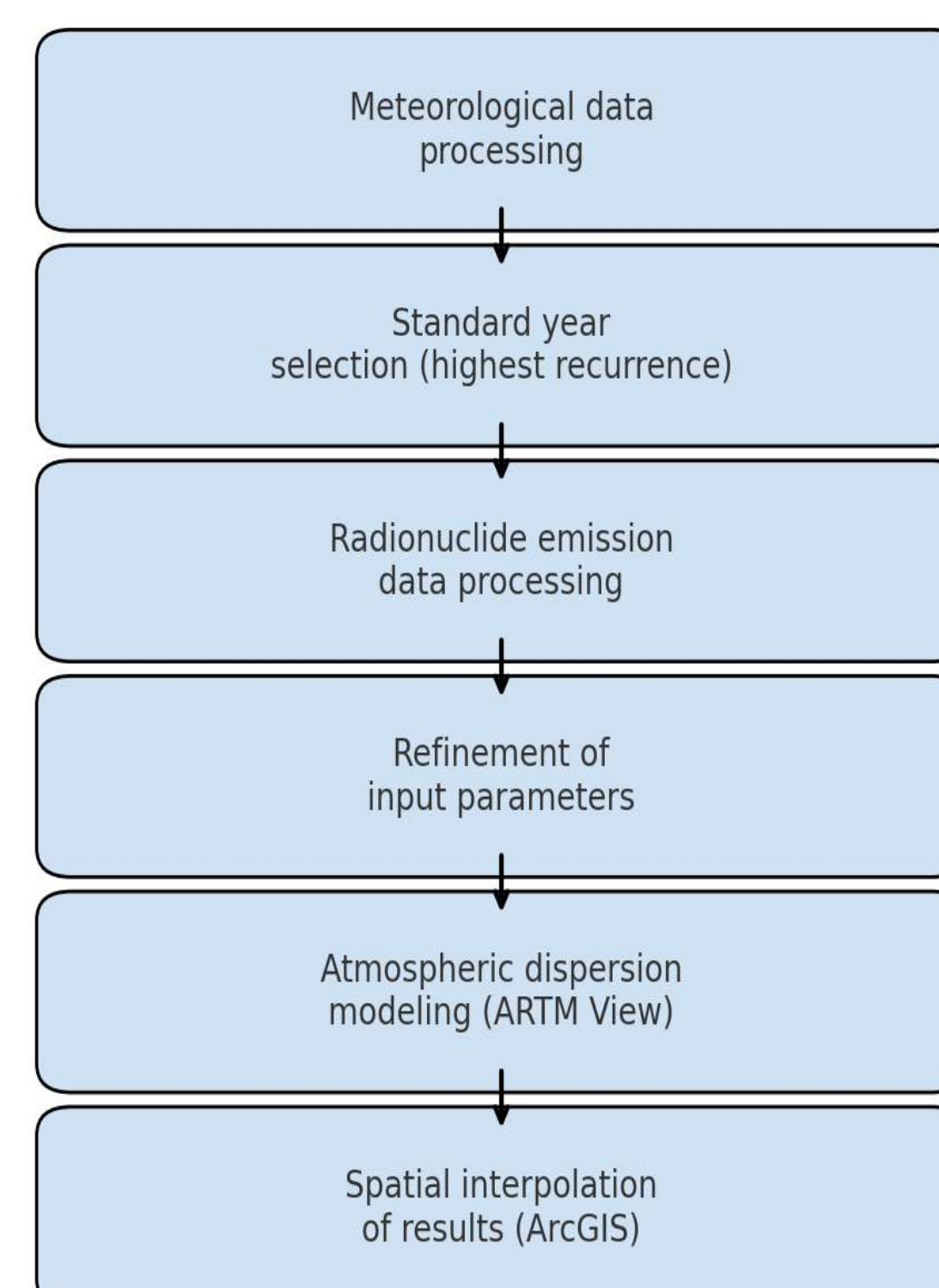
INTRODUCTION

The Radiopharmaceutical Production and Research Unit (UPPR) of the Nuclear Technology Development Center (CDTN/CNEN, Brazil) plays a key role in supplying radioisotopes for medical applications. Ensuring that atmospheric releases of radionuclides remain within safe limits is therefore critical not only for regulatory compliance but also for protecting public health and the environment. In this study, radionuclide dispersion in the atmosphere was assessed in accordance with CNEN Standard NN 3.01, using the internationally recognized ARTM View model developed by the German Federal Office for Radiation Protection (BfS).

MATERIALS AND METHODS

- Meteorological data: a robust 20-year dataset (1997–2017) of locally recorded weather conditions.
- Radionuclide emissions: continuous releases of Fluorine-18, Nitrogen-13, and Carbon-11.
- Environmental parameters: detailed characterization of topography, terrain roughness, physical obstacles, stack height, and wind fields.
- Modeling tools: ARTM View for atmospheric dispersion simulations and ArcGIS for spatial representation of results

Methodology Flowchart



OBJECTIVES

- ✓ To model the atmospheric dispersion of the radionuclides Fluorine-18, Nitrogen-13, and Carbon-11 released from the facility.
- ✓ To estimate the annual effective dose to the identified critical groups of the surrounding population, ensuring compliance with regulatory limits.
- ✓ To generate spatial representations of radionuclide concentration, ground deposition, and dose through thematic mapping.
- ✓ To provide technical support for decision-making regarding the design and placement of environmental monitoring systems.

ACKNOWLEDGEMENTS

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RESULTS

Dispersion pattern: Radioactive plumes drift predominantly northwest, consistent with prevailing winds.

Maximum ground-level concentrations:

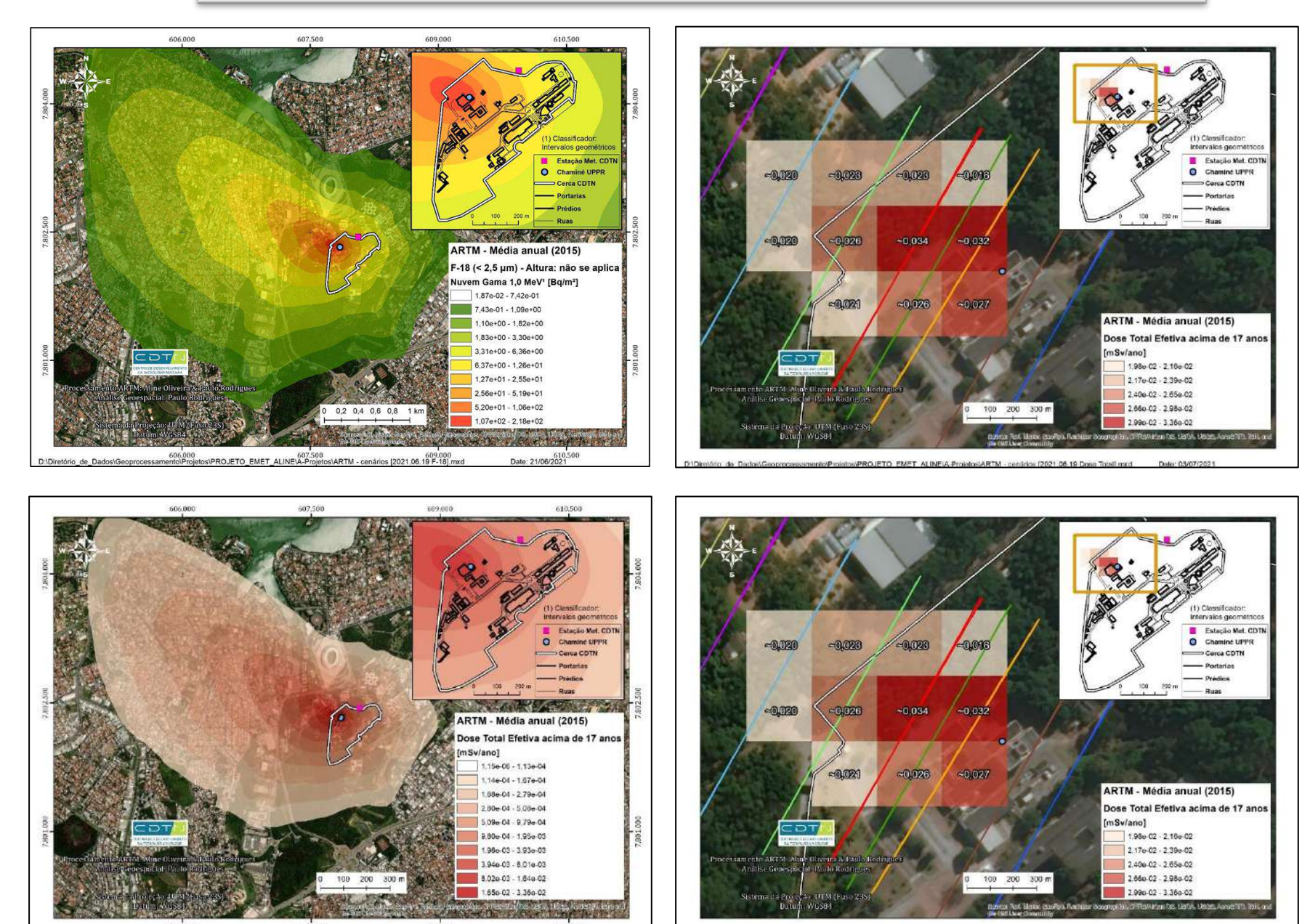
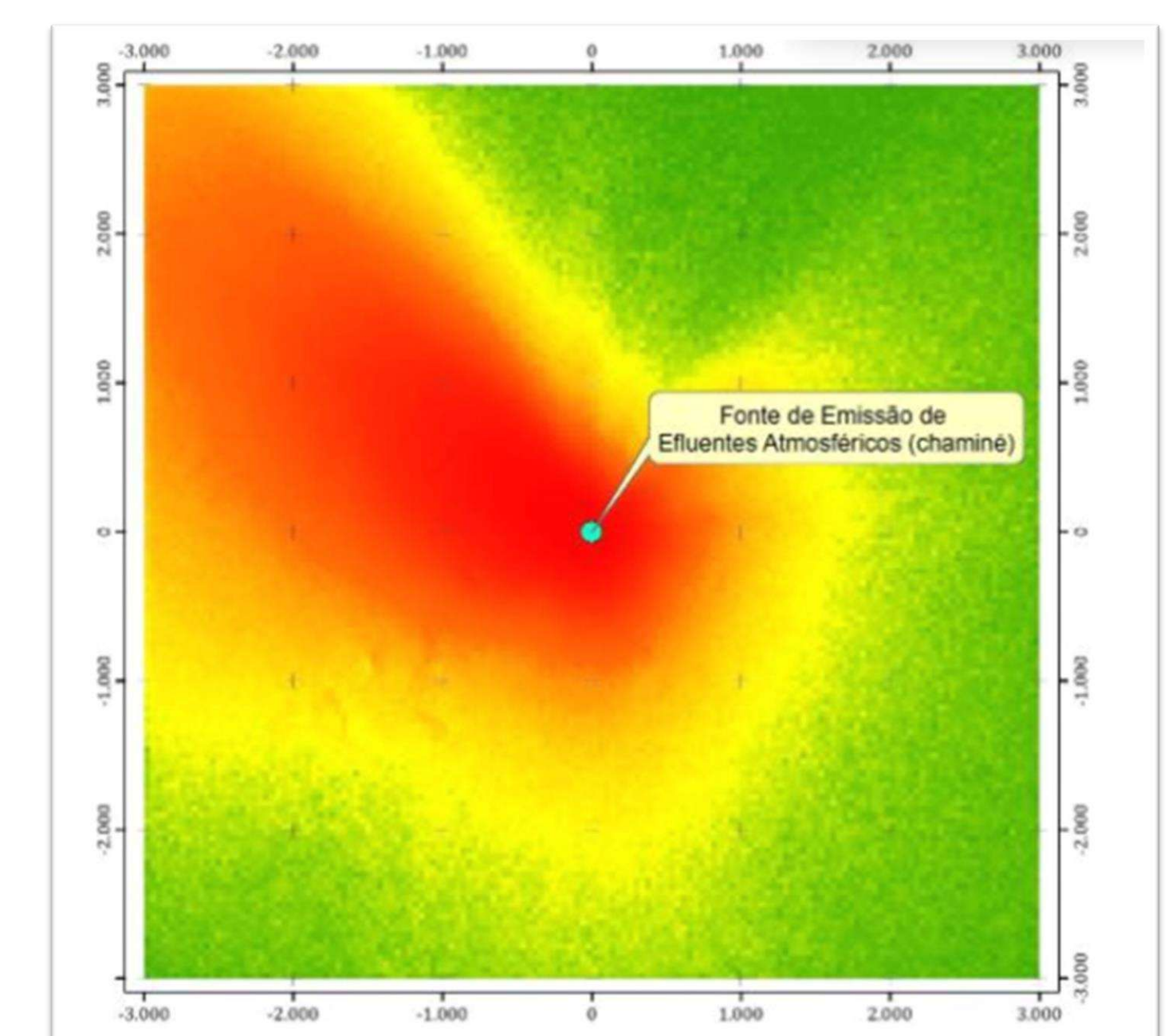
- F-18 → 5.87 Bq/m³
- N-13 → 0.45 Bq/m³
- C-11 → 1.62 Bq/m³

Regulatory compliance: All concentrations remained well below CNEN release limits.

Public exposure:

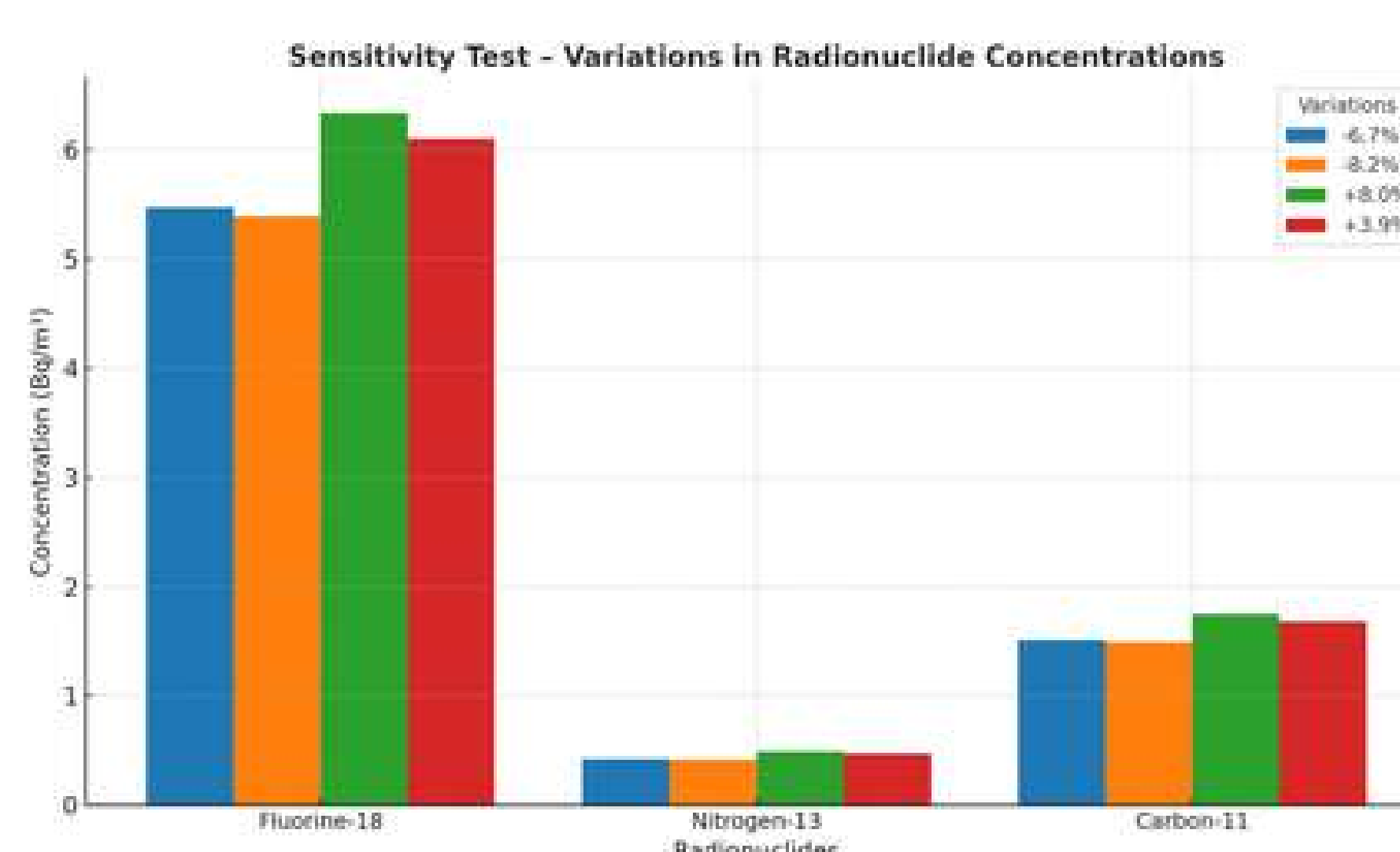
Estimated **annual effective dose** = 0.033 mSv/year, nearly **10 times lower** than the reporting threshold (**0.3 mSv/year**).

Dados gerais	Faixa Etária				
	< 2 anos	2 a 7 anos	7 a 12	12 a 17 anos	> 17 anos
DOSE FINAL MÁXIMA (mSv/ano)	2,14E-01	2,14E-01	1,71E-01	1,48E-01	1,44E-01



SENSITIVITY TESTS

Simulations were performed to evaluate the influence of key input variables on dispersion outcomes. Results indicated that source height and wind speed were the most critical factors, significantly affecting plume direction and concentration patterns. These findings highlight the need for precise parameterization and model calibration to ensure reliable assessments.



CONCLUSIONS

The study demonstrated that, when supplied with refined input data and operated by a qualified team, the ARTM View model is a reliable and effective tool for assessing radionuclide atmospheric dispersion. The results confirmed that emissions from CDTN/UPPR remain well within regulatory limits, ensuring the safety of surrounding populations and compliance with CNEN standards. Moreover, the methodology and findings are transferable to other nuclear facilities, reinforcing ARTM View as a valuable resource for environmental monitoring and radiological protection.